

NITHIN KATLA

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PROFESSIONAL EXPERIENCE

Research Assistant – Visiting Scholar

Jul. 2025 – Present

Stony Brook University – Visitor Scholars & Interns Program

Seattle, WA

- Co-led the design of a distributed consensus orchestrator in Go, applying advanced data structures and algorithms to benchmark RCC, DQBFT, MirBFT and Dispel protocols under varied workloads for scalability and resilience.
- Integrated a reinforcement learning agent with the orchestrator to dynamically switch protocols, improving throughput by 25% while preserving safety guarantees in Byzantine fault environments.
- Implemented full-stack observability with Prometheus and Grafana to monitor latency, throughput and recovery metrics, enabling proactive detection of bottlenecks and faster incident resolution.

Graduate Research Assistant & Lead Developer

Dec. 2023 – Jan. 2025

Stony Brook University – Mechanismic Inc.

Stony Brook, NY

- Designed and launched a browser-based 3D robotics simulation platform using React.js and Babylon.js, modernizing legacy code into a scalable, cloud-ready platform.
- Optimized Redux Toolkit state management and IndexedDB caching, reducing load times by 30% and improving data retrieval efficiency for large simulation models.
- Architected and deployed Go microservices with Docker and Kubernetes, provisioning infrastructure via Terraform for resilient, cloud-native deployment.
- Built Kafka-based telemetry pipelines with Prometheus and Grafana monitoring to ensure 99.9% uptime, addressing data integrity and reliability for real-time streaming workloads.
- Authored design documentation and API specifications (REST, GraphQL), improving developer experience and system maintainability.

Specialist Programmer (Full Stack Engineer)

Sep. 2021 – Aug. 2023

Apple Inc. – Contract via Infosys Ltd

Hyderabad, India

- Developed interactive React.js dashboards to visualize and modernize Apple Maps data ingestion workflows, reducing vendor discrepancies by 70%.
- Engineered backend REST APIs with Java Spring Boot and Oracle DB, designing efficient schemas and optimizing SQL queries for high-performance transactional analytics.
- Built and optimized Spark jobs on AWS EMR to crawl vendor sites and ingest 100M+ POI records/day into Apple Maps, with schema validation, rate limiting, and skew handling for correctness and scalability.
- Improved API responsiveness from 400ms to 150ms by integrating Redis caching and tuning database queries, strengthening overall system performance and resiliency.
- Participated in cross-functional code reviews and architecture discussions, ensuring high availability, modernization and adherence to best engineering practices.

Software Engineer

Apr. 2020 – Mar. 2021

Dhan Technology Labs India Private Limited

Hyderabad, India

- Built AI-driven conversational platforms with Rasa and React.js, leveraging NLP pipelines to achieve 90% intent accuracy and reducing response latency by 30%.
- Implemented backend APIs in .NET Core with SQL Server and Redis caching, improving throughput and optimizing relational schema designs for performance.
- Contributed to modernization of web applications with HTML5, CSS3 and JavaScript ES6, improving scalability, usability and responsiveness across devices.
- Wrote unit and integration tests (Jest, Selenium) to ensure code quality, reliability and smooth deployment in production environments.

TECHNICAL SKILLS

Languages & Frameworks: Python (Django), JavaScript/TypeScript (React.js, Next.js, Node.js), Java (Spring Boot, Concurrency, Memory Management), Go, Rust, HTML5/CSS3

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

APIs, Messaging & Streaming: REST API, GraphQL, gRPC, OAuth2/JWT, Apache Kafka, RabbitMQ, API Gateway (AWS, Kong)

Cloud, DevOps & CI/CD: AWS, GCP, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Webpack

Testing, Distributed Systems & Observability: Jest, Cypress, Selenium, React Testing Library, Quota & Rate Limiting, Tenant Isolation, Traffic & Load Balancing, Prometheus, Grafana

AI/ML: TensorFlow (RL), scikit-learn, OpenAI GPT, Prompt Engineering, Pandas, Numpy, Deep Learning(CNN)

PROJECTS

Fault-Tolerant Replication Protocol with PBFT | *Distributed Systems, Go, gRPC, Protobuf*

- Designed and implemented a PBFT-based replication protocol using threshold signatures, checkpointing, and garbage collection, ensuring linearizability with **linear communication complexity**. Optimized for high concurrency, processing up to **10,000 client requests per second** under Byzantine fault tolerance.

Distributed Transaction Processing System with Paxos | *Distributed Systems, Go, gRPC, Protobuf*

- Architected and developed a fault-tolerant banking system for intra and cross-shard transactions using sharding and a modified Paxos protocol with **two-phase commit** and locking. Ensured **eventual consistency** and high reliability through write-ahead logging (WAL) and transaction ordering, achieving **99.99% uptime** in testing.

EDUCATION

Stony Brook University

Aug 2023 – May 2025

Master of Science in Computer Science

Stony Brook, NY, USA

Relevant Courses: Distributed Systems, Operating Systems, Data Visualization, Machine Learning, Theory of Database Systems

Jawaharlal Nehru Technological University

June 2017 – June 2021

Bachelors in Computer Science

Hyderabad, India

Relevant Courses: Cloud Computing, OS, Computer Networks, Software Engineering, User Experience Engineering, Cyber Security